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Benchmarking M6 competitors: An analysis of financial metrics and discussion of incentives☆

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ABSTRACT

The M6 Competition assessed the performance of competitors using a ranked probability score and an information ratio. While these metrics do well at picking the winners in the competition, crucial questions remain for investors with longer-term incentives. To address these questions, we compare the competitors' performance with a number of conventional (long-only) and alternative indices using industry-relevant metrics. We apply factor models to measure the competitors' value-adds above industry-standard benchmarks and find that competitors with more extreme performance are less dependent on the benchmarks. We further introduce two new strategies by picking the competitors with the best (Superstars) and worst (Superlosers) recent performance and show that it is challenging to identify skill amongst investment managers. Finally, we discuss the incentives of winning the competition compared with professional investors, where investors wish to maximize fees over an extended period of time, and provide suggestions for future competition improvements.

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1. Introduction

1.1. Benchmark indices and net asset values

By the end of the M6 Competition on 3 February 2023, 58% of competitors had realized negative returns. The equal-weighted benchmark was up 0.5% and consisted of 50 stocks and 50 exchange-traded funds (ETFs). Of the 163 teams that completed all 12 months of the M6 Competition, only 28% had a realized return higher than that of the equal-weighted benchmark, demonstrating the difficulty of the competition and reflecting the challenges all active managers face in beating broad benchmarks. On the other hand, the competition does not fully capture the scope of running an active investment management offering, in the following sense: professional investors ostensibly wish to maximize fees over a long period of time, while a short-term competition may encourage riskier behavior, as the upside of winning the competition dwarfs the potential downside of not winning, and may induce the competitors to engage in less than entirely rational behavior (Bali et al., 2011). To be clear, this is not a suggestion that the competition is flawed in any way or that all competitors performed extremely, but simply an attempt to recognize that the incentives of the competition are somewhat different from those of running a hedge fund and, as such, may encourage different, possibly riskier behavior (Lichtendahl & Winkler, 2007; Witkowski et al., 2023).

Our contributions in the paper are threefold. First, we document the performance of M6 competitors empirically using a range of industry-relevant financial metrics. In contrast to Makridakis et al. (2024), we introduce

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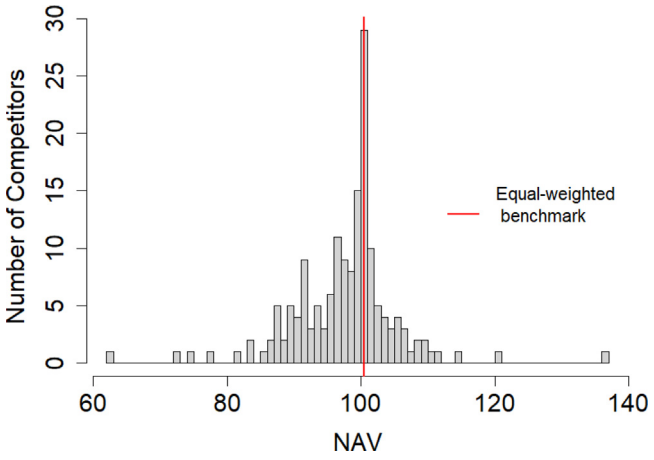


Fig. 1. Histogram of the ending NAVs for all competitors when they started the competition with $100.

new metrics that competitors were not rewarded for, to better understand the performance of competitors from a practice perspective. Second, we apply factor models to competitors' returns and find that competitors who performed at the extremes tended to be less dependent on the benchmark indices. Specifically, we regress each competitor's returns against the returns of seven benchmark indices to calculate the percentage of variation in a competitor's returns that benchmark indices can explain. We find that when the percentage of variation in a competitor's returns explained by benchmark indices is high, the competitors are not likely to have outstanding performance in the competition. To judge competitors' performance, we use the regression intercept (α), which demonstrates a competitor's independent value add. Conclusions are similar when we use the Sharpe ratio or information ratio in place of the regression intercept. Third, we test the momentum of competitors using two new strategies (Superstars and Superlosers) and find it challenging to identify repeatable skill amongst competitors.

Fig. 1 shows a histogram of the net asset values (NAVs) among competitors, representing the dollar value at the close of the competition on 3 February 2023. It assumes that each competitor started with $100 at the start of the competition. Notably, only four teams ended with a NAV less than $80 (at least a 20% loss) and only two teams ended with a NAV greater than $120 (at least a 20% gain). Furthermore, 68 (42%) competitors increased their investments during the competition with a NAV exceeding $100, 95 (58%) decreased their investments, and 15 (9%) competitors had daily returns and NAVs identical to those of the equal-weighted benchmark, as illustrated by the large spike in the middle.

A reasonable starting point for considering relative performance is the S&P 500 and Nasdaq-100, two large-cap indices representing large-cap U.S. stocks, which many passive investors track via low-cost exchange-traded funds (ETFs). The majority of assets in the universe for the competition are individual large-cap equities or ETFs that track some set of large-cap equities. The performance of the competitors relative to these passive indices was encouraging, as 2022 was a very challenging year for equities, and both indices experienced significant drawdowns during the timeframe of the competition. This suggests that while most competitors did not add value over the equal-weighted portfolio of universe constituents, a majority did manage to express smaller losses than widely used long-only benchmarks. This highlights one of the arguments made for active investment management and the M6 competition: Active management can reduce risk, independent of attempting to generate market-beating returns. Notably, a large part of the active management was the M6 organizers themselves – who pre-filtered the universe of stocks and ETFs from a clustering solution of average stock price, coefficient of variation, standard deviation, volume, and total returns (Makridakis et al., 2024). This appeared to be a successful strategy for outperforming other benchmark indices using an equal-weighted benchmark. Another point to note is that, as discussed in DeMiguel et al. (2009), it is extremely difficult to beat a simple equal-weighted allocation with more complex methods.

As the competition was related to active management and allowed competitors to use both long and short positions, it also makes sense to look at active benchmarks. When the NAVs of both large-cap and active benchmark indices[1] during this same time period are considered, the S&P500 ends with a NAV of $95.56, NASDAQ-100 with $90.86, Crypto with $62.54, Commodity Trading Advisors (CTA) with $104.26, Equity Long–Short (LS) with $99.99, Event-Driven with $101.33, and Market Neutral with $101.93. A more optimistic look emerges when competitors are compared with the ending NAV of the S&P 500, showing that only 50 competitors (31%) had a lower ending NAV. On the average, the M6 competitors ended with a NAV of $97.60 and the equal-weighted benchmark ended with a NAV of $100.53.

However, besides the ending NAV, the pathways to the ending NAV are usually a major focal point for investors.

[1] Data were made available to us from https://nilssonhedge.com/index/.

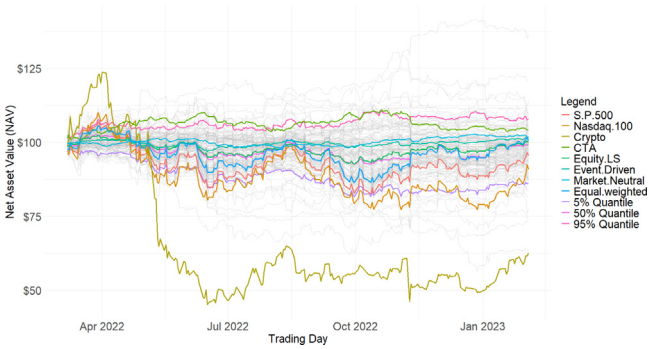


Fig. 2. Net asset values of all competitors, benchmark indices, and quantiles for each day of the M6 Competition.

The NAV can be measured at the close of each trading day as

$$NAV_t = \$100 \times \prod_{t'=1}^{t'=t} (1 + r_{t'}), \tag{1}$$

where t is the trading day and $r_{t'}$ is the percent return on trading day t'. In this paper, we use raw returns, r_t, which differ from the log returns used by Makridakis et al. (2024), $r_t^* = \log(1 + r_t)$, but are a close approximation unless the returns are extreme. To transform log returns to raw returns, we use the equation $r_t = \exp\left(r_t^*\right) - 1$.

Fig. 2 plots the daily NAVs of the M6 competition for all competitors, benchmark indices, and three quantiles (5%, 50%, 95%) of competitors. The quantile NAVs are calculated as the q% quantile of all 163 competitor NAVs on day t. They do not track a single competitor across time. The figure shows that the Crypto index and 16 (10%) competitors are outside the 5% and 95% quantiles, demonstrating more extreme return streams. Most of the other benchmark indices are within the quantiles, except for the Nasdaq-100, which exceeded the 95% quantile early in the competition and the 5% quantile several times later in the competition. A savvy investor may also question whether the top 5% (or bottom 5%) of competitors exhibit mean reversion behavior – that is, does a large gain in the past month lead to a large loss in the next month? We investigate this phenomenon in Section 3 based on a well-known momentum study (Burghardt et al., 2007), which shows that investing in past superstars may not be optimal.

1.2. Financial metrics

Investors consider a variety of financial metrics when making investment decisions. The most prominent metric is the annualized Sharpe ratio (SR), while the M6 Competition (Makridakis et al., 2024) uses the related information ratio (IR),

$$IR = \frac{\sum_{t=1}^{t=n} \log(1 + r_t)}{\sqrt{\frac{1}{n-1}\sum_{t=1}^{t=n}(\log(1 + r_t) - \frac{1}{n}\sum_{t=1}^{t=n}\log(1 + r_t))^2}} \tag{2}$$

where r_t is a portfolio's raw percent return at the close of day t and n is the total number of trading days (238).

The annualized SR is defined as

$$SR = \frac{\prod_{t=1}^{t=n}(1 + r_t)^{n^*/n} - 1}{\sqrt{n^*\sigma}}, \tag{3}$$

where t is the trading date, n^* is the total number of periods in a year (252), and σ is the standard deviation of daily returns. The numerator of Eq. (3) is the annualized return (without the log approximation), and the denominator is the annualized standard deviation. As a result, the IR can be over an order of magnitude higher than the SR, but the correlation should be very close to 1 with reasonable returns (if the raw returns are unreasonably negative or positive, e.g., 30% or −30%, then the log approximation can break down, resulting in returns of 26.2% and −35.7%, respectively).

The primary difference between the M6 Competition's use of the IR in Eq. (2) and our use of the SR in Eq. (3) is that we annualize the expected raw returns and standard deviations to compute the SR, while the M6 Competition uses log returns and does not annualize the standard deviation, resulting in higher IR values (the correlation between SR and IR across competitors is 0.996 with only slight differences due to the log approximation). Sharpe (1994) suggests annualizing and compounding multi-period returns when calculating the SR. Regardless of IR or SR, investors still expect a positive value but desire SRs to be closer to 2. SRs exceeding 2 are more theoretically ideal but problematic, because they may indicate fraud or trading strategies that miraculously fail at one point in the future. For example, one of Bernie Madoff's feeder funds, the Fairfield Sentry Ltd. hedge fund, had an annualized SR of 2.47 from December 1990 to October 2008 (Bernard & Boyle, 2009). Also, portfolios that short put options on the S&P 500 can have an unusually high SR until a market downturn. Many portfolio managers deem these strategies as "too good to be true" and avoid them due to perceived risks, even though the past data may never have shown a large downturn.

Investors also like to look at a statistic called the maximum drawdown (MDD), which is the maximum percentage loss over the entire history of the portfolio, from peak to trough. For example, if Portfolio A started with $100 and had the consecutive daily NAVs $110, $200, $100, $150, the MDD would be 50% because the maximum percentage loss occurred from $200 to $100, representing a 50% decrease. However, if Portfolio B started with $100

and had the consecutive daily NAVs \$95, \$90, \$85, \$80, the MDD would be 20%. An investor might prefer Portfolio B due to the lower MDD, even though Portfolio A had a higher return than Portfolio B. The MDD is defined as

$$MDD = \max_{t_1<t_2} \frac{-\left(NAV_{t_2} - NAV_{t_1}\right)}{NAV_{t_1}}, \tag{4}$$

where $t_1 = 1, \ldots, n$ and $t_2 = 2, \ldots, n$ for all $t_1 < t_2$. Since the MDD is highly dependent on time, it depends on when an investor enters a portfolio. Thus, two different investors can have completely different MDDs for the exact same portfolio. Although MDD is a maximum-order statistic lacking in statistical theory (see Section 2.3 for the theoretical explanation), it influences outcomes and investor decisions. For example, investors commonly issue stop losses at an MDD threshold to reduce their overall risk.

Investors can replace the denominator of the SR in Eq. (3) with the MDD in Eq. (4) to create the Calmar ratio (CR), but this metric suffers from similar maximum order statistic problems. In the case of Portfolio A, which has a higher standard deviation of returns and MDD than Portfolio B, decisions may be similar. The signs of the SR and CR are entirely dependent on the expected returns, as they share the same numerator. The CR is defined as

$$CR = \frac{\prod_{t=1}^{t=n}(1+r_t)^{n^*/n} - 1}{MDD}, \tag{5}$$

where the annualized return is the same as the numerator in Eq. (3).

Investors are also concerned about the duration of any drawdown until recovery to the prior peak, producing more interesting metrics, such as the ulcer index (UI; Martin & McCann, 1989) and the ulcer performance index (UPI). The UI measures the pain (duration and depth) of investor losses, from peak to trough to peak of a portfolio's drawdowns,

$$UI = \sqrt{\frac{1}{n}\sum_{t=1}^{t=n} DD_t^2}, \tag{6}$$

where DD_t is the drawdown on trading day t since the previous peak. UI is a practical metric because investors are more likely to withdraw their assets the longer a portfolio is underwater. For example, the UI of Portfolio C with NAVs of \$100, \$90, \$100, \$110 would be 0.058 (with drawdowns of 10%, 0%, and 0%), while the UI of Portfolio D with NAVs of \$100, \$90, \$90, \$110 would be 0.082 (with drawdowns of 10%, 10%, and 0%), representing a greater ulcer for an investor in Portfolio D. Last, the UPI replaces the denominator of the SR in Eq. (3) with the UI in Eq. (6),

$$UPI = \frac{\prod_{t=1}^{t=n}(1+r_t)^{n^*/n} - 1}{UI}. \tag{7}$$

Not all volatility is bad. For example, two portfolios may have identical negative returns but different positive returns. In this case, the portfolio with the larger positive returns has a higher standard deviation, which any investor prefers. To account for this, we measure each portfolio's upside volatility (vol^+) and its downside volatility (vol^-). Upside volatility is defined as

$$vol^+ = \sqrt{\frac{\sum_{t=1}^{t=n_1}\left(r_t^+ - \frac{\sum_{t=1}^{t=n_1} r_t^+}{n_1}\right)^2}{n_1}}, \tag{8}$$

where r_t^+ denotes the positive raw returns and n_1 is the sample size of the positive returns. Similarly, downside volatility is defined as

$$vol^- = \sqrt{\frac{\sum_{t=1}^{t=n_2}\left(r_t^- - \frac{\sum_{t=1}^{t=n_2} r_t^-}{n_2}\right)^2}{n_2}}, \tag{9}$$

where r_t^- denotes the negative raw returns and n_2 is the sample size of the negative returns.

Note that the case of $vol^+ \gg vol^-$ is not accounted for in the denominator of the SR or IR, but it is more likely to occur when the numerator is positive. Investors prefer $vol^+ \gg vol^-$, and in such a case, the MDD is likely to be smaller.

Table 1 presents the financial metrics for the major benchmark indices, including the equal-weighted benchmark in the M6 Competition.[2] Results are presented in increasing order of IR and SR (with a risk-free rate of 0) over the $n = 238$ trading days of the competition. SRs were relatively moderate (between -1 and 1), and SRs were positive for those indices that gained money ($NAV_{238} >$ \$100). The relationship between the SR and CR was also monotonic for the benchmark indices, indicating that the MDDs were monotonically increasing with the annualized standard deviation of returns. MDD and UI were also monotonically increasing with each other, indicating that the lengths of the drawdowns were not greatly mismatched to the depths of the drawdowns.

Table 2 shows that most competitors had a negative SR (with NAVs below \$100), and the SRs were between -1.76 and 2.58. Furthermore, some competitors had high UI values, illustrated by some of the gray NAV lines in the bottom part of Fig. 2. The values of first-order autocorrelation seemed to be moderate, but some competitors had high MDDs.

1.3. A comparison of the information ratio with alternative risk-adjusted metrics

The goal of the M6 competitors was to maximize their IR to win the competition. We explore whether achieving this goal extends to the other financial metrics that are also based on return-to-risk ratios. The comparable financial metrics include the SR, CR, and UPI and two new risk-adjusted metrics (adjusted SR and modified SR are introduced below).

[2] For the analysis in this paper, we removed the Crypto returns that occurred on the weekends (non-trading days) and assumed that all benchmark indices and competitors traded on the same 238 trading days for a fair comparison. Crypto had an average daily return of -0.115% when all 334 days that spanned the competition were included, and an average daily return of -0.126% when only the 238 trading days were included.

Table 1
Financial metrics of the major benchmark indices.

	IR	SR	MDD	Autocorrelation	vol^+	vol^-	CR	UI	UPI
Crypto	−12.32	−0.66	0.64	−0.04	0.02	0.03	−0.62	0.63	−0.62
Nasdaq-100	−4.77	−0.30	0.30	−0.05	0.01	0.01	−0.32	0.21	−0.46
S&P 500	−3.02	−0.20	0.23	−0.01	0.01	0.01	−0.21	0.14	−0.33
Equity LS	−0.01	0.00	0.10	−0.02	0.00	0.00	0.00	0.05	0.00
Equal-weighted	0.45	0.03	0.18	0.04	0.01	0.01	0.03	0.10	0.06
Event Driven	5.40	0.36	0.04	0.05	0.00	0.00	0.35	0.02	0.92
Market Neutral	9.18	0.62	0.03	0.17	0.00	0.00	0.64	0.02	1.34
CTA	9.32	0.64	0.07	0.18	0.00	0.00	0.69	0.04	1.28

Table 2
Summary statistics of the financial metrics for M6 competitors.

	IR	SR	MDD	Autocorrelation	vol^+	vol^-	CR	UI	UPI
Min.	−29.18	−1.76	0.01	−0.13	0.00	0.00	−1.00	0.01	−2.44
1st Qu.	−8.37	−0.53	0.10	0.00	0.00	0.00	−0.47	0.05	−0.81
Median	−1.34	−0.09	0.16	0.04	0.01	0.01	−0.11	0.08	−0.22
Mean	−3.06	−0.18	0.15	0.02	0.01	0.01	−0.04	0.08	0.04
3rd Qu.	1.02	0.07	0.18	0.05	0.01	0.01	0.07	0.10	0.13
Max.	32.89	2.58	0.47	0.16	0.02	0.02	4.95	0.36	11.91

From a winner's perspective, the rank order of the IR and SR is the same for the top 10 competitors, while the CR and UPI have a few differences but keep the same 10 competitors within the top 10 places. In contrast to the M6 competition's IR, both the CR and UPI switch the places of the second and third place competitors, indicating that competitors minimizing standard deviations of returns (denominator of IR) may have a slightly different result than competitors minimizing maximum drawdowns (denominator of CR).

For an association measure, the Pearson correlation between the ranks of two financial metrics is equivalent to the Spearman correlation. A perfect value of 1 indicates that the competitors are perfectly aligned in terms of their performance using either metric—for example, the 143rd placed competitor in the IR is still the 143rd placed competitor in another financial metric. The Spearman correlation for the SR is 0.9998, for the CR is 0.9933, and for the UPI is 0.9795. Next, we investigate alternative risk-adjusted metrics by changing the assumptions of risk in the denominator of the SR.

The first risk-adjusted metric, the adjusted SR (aSR), adjusts the standard deviation of the SR in Eq. (3) for autocorrelation. The problem with annualizing the standard deviation is that the returns may not be independent. The SR formula in Eq. (3) for annualizing the standard deviation in the denominator assumes independence by scaling the daily standard deviation by $\sqrt{n^*}$. For aSR, we adjust the denominator of the SR by incorporating the autocorrelation coefficients per Burghardt and Liu (2012) as follows,

$$aSR = \frac{\prod_{t=1}^{t=n}(1+r_t)^{n^*/n} - 1}{\sqrt{n^*}\sigma\sqrt{(1+2\sum_{i=1}^{i=k}\rho_i)}}, \tag{10}$$

where ρ_i is the ith order autocorrelation up into k lags. The basic idea is that the aSR should be higher if the daily returns are, on the average, negatively autocorrelated ($\sum_{i=1}^{i=k}\rho_i < 0$). As a result, the annualized standard deviation should be lower in the presence of negative autocorrelation, since the negatively correlated returns cancel each other out across 252 time periods. On the other hand, the annualized standard deviation should be higher in the presence of positive autocorrelation. If the daily returns are independent, there should be no or minimal effect (0 autocorrelation).

The second risk-adjusted metric, the modified SR (mSR), modifies the denominator of the SR to a p% value-at-risk (VaR). The p% VaR represents the loss that's exceeded on the worst p% of days. For example, if a competitor realized a 3% or greater loss on 5% of their trading days, then the 5% VaR would be equal to 3%. A 1% VaR would obviously be a larger loss on the same return data. For reproducibility and comparison with the other financial metrics in the paper, we use the PerformanceAnalytics package in R for the annualized mSR. Otherwise, the calculation is fairly straightforward using a Cornish–Fisher expansion (Favre & Galeano, 2002), which requires calculating skewness, kurtosis, standard deviation of returns, and a z-score based on the p% quantile.

Table 3 displays the financial metrics for the top five competitors in the M6 Competition based on their values of IR. All top five competitors in IR were also in the top five for each financial metric. The results show that the first place winner would have changed if the aSR was used, because it rewards negative autocorrelation (lower risk) and penalizes positive autocorrelation (higher risk). The deviations from SR to aSR can be seen by comparing the values in the table across the rows. When the aSR is greater than SR, it implies that the first 10 lags of autocorrelation were negative, on the average. In addition, the second place winner would have changed if a drawdown-based risk was used (CR and UPI) or a 5% VaR was used (mSR). However, it is important to note that although there might be advantages to a negatively autocorrelated portfolio in practice, the competition incentivized IR. The results for these other financial metrics could have changed if competitors optimized their loss functions on negative autocorrelations, drawdowns, or VaRs.

As a measure of concordance, we use Kendall's W, which measures how differently the financial metrics

Table 3
Top five competitors for each risk-adjusted metric.

Competitor rank	IR	SR	CR	UPI	aSR	mSR (5% VaR)	mSR (1% VaR)
1	**32.89**	**2.58**	**4.95**	**11.91**	1.99	**36.46**	**16.80**
2	28.66	2.11	3.84	9.76	2.42	22.95	14.80
3	24.72	1.72	4.22	10.16	**2.88**	20.49	12.55
4	21.54	1.52	2.55	5.64	1.74	23.88	6.37
5	17.94	1.23	2.32	6.04	1.19	19.56	3.31

Note: First place is **bold** and second place is underlined.

rank the performance of competitors. A value of 1 indicates that all financial metrics unanimously agree, while a value of 0 demonstrates randomness. For all seven metrics in Table 3, the Kendall's W statistic is 0.990 with a p-value less than 0.001 to reject the null hypothesis that Kendall's W is 0. Overall, the results were fairly robust when the competition's metric (IR) was compared with other risk-adjusted metrics.

1.4. Management and performance fees

Another primary consideration for investors is the effect of management and performance fees on portfolio performance. Management fees are annualized flat percentage fees collected off the annual NAV. Hedge funds collect these monthly or quarterly, typically between 1% and 2% annually. Performance fees are incentive fees collected on the increase in NAV from an initial NAV, typically between 10% and 20%. Elton et al. (2003) found that funds with incentive (performance) fees have higher average risk-adjusted performance than funds without incentive fees, which suggests that they tend to attract superior managers. In the context of the design of the M6 Competition, the competitors did not charge management or performance fees, but we analyzed the effect of moderate (1%/10%) fees.

Management and performance fees are difficult to calculate, since the performance fee is only incurred on new high-water marks (HWMs). For example, if Portfolio A went from a \$100 NAV to a \$200 NAV (new HWM), there would be a 10% performance fee on the gain (+\$100) for a total of \$10, reducing the new NAV from \$200 to \$190. However, if portfolio A had a subsequent drawdown, the performance fee would not be incurred until a new NAV exceeded the previous HWM of \$200. For this reason, the performance fee calculation is highly dependent on when an investor invests and is usually outsourced to an accounting firm. For simplicity, with the short timeline of the M6 Competition, we assume a low fee structure (1%/10%) and deduct performance fees and prorated management fees daily.

Fig. 3 plots the 5%, 50%, and 95% quantiles of the NAVs for all 163 competitors in the M6 competition. NAVs are shown before fees (blue, dark blue, and black for the 5%, 50%, and 95% quantiles, respectively) and after fees (green) by assuming 1% management and 10% performance fees. The difference between non-fee and fee portfolios heightens as the competition length increases. Since the competition lasted for only 238 trading days, the net effect of the 1%/10% fees were not substantial. For the median competitor, the 1%/10% fees decreased the SR from −0.09 to −0.17, the IR decreased from −1.3 to −2.6, and the MDD increased from 15.5% to 16.0%. However, if the competition lasted several years, the effects of management and performance fees would be much more severe. Unlike the short-term nature of the competition, the portfolio managers' goal is to retain assets under management (AUM) and maximize these fees over a long period of time.

2. Statistical analysis of the M6 competitors

2.1. Statistical analysis of a competitor's value-add

The statistical performance is assessed by measuring the value-add (α) of competitors' portfolios compared with major indices. Alpha is frequently mentioned on a portfolio's marketing sheet and represents the independent return that a portfolio produces above and beyond benchmark indices.

From a statistical perspective, α is the estimated intercept of a multiple linear regression (or factor model). It represents the excess return over the explanatory power of the predictor (X) variables. The predictor variables are typically benchmark indices or other factors found in the suite of Fama–French regression models (Fama & French, 1995). Intuitively, α represents the additional return to a portfolio when all benchmark indices produce a 0 return. Investors may compare benchmark indices with minimal fees, such as the S&P 500 or the Nasdaq-100, with the focal portfolio. The idea is that investors would not want to pay management and performance fees if a portfolio cannot produce excess returns independent of the minimal fee benchmark indices. The standard approach to identify which portfolios have positive α is "to perform many individual statistical tests on the alphas of the factors or funds relative to the benchmark model, and then make a selection based on the significance of these individual tests" (Giglio et al., 2021, p. 3457).

For the factor model, let r_t be the daily return of a competitor's portfolio on day t, r_t^f be the de-annualized risk-free rate at day t, α be the estimated intercept of a multiple linear regression, $f_{j,t}$ be the daily return of a benchmark index (factor) on day t, β_j be its coefficient, and ε_t be the standard error term that is independent and identically distributed (i.i.d.) with mean 0 and constant variance from a normal distribution,

$$r_t - r_t^f = \alpha + \sum_{j=j}^{j=J} \beta_j \left(f_{j,t} - r_t^f \right) + \varepsilon_t. \tag{11}$$

For a portfolio manager to market their portfolio effectively, α should be positive. This desired criterion creates



Fig. 3. The effect of management and performance fees on competitor NAVs.

a selection bias for investors because portfolio managers rarely advertise portfolios with negative α. Furthermore, portfolio managers may introduce additional biases to present a more favorable α, such as changing the benchmark indices, altering the time windows, reducing management and performance fees, or ignoring the lack of statistical significance. A major advantage of the M6 Competition structure is that most of these biases can be controlled for using the same benchmark indices, fees, time windows, and statistical significance calculations. In addition, none of the data are missing, which would tend to pose problems for the computation of α (Giglio et al., 2021).

The appraisal rRatio (AR; Hecht, 2014), which divides a portfolio's α by the standard deviation of the factor model's residuals, is less known. The AR is a risk-adjusted measure of α and is shown below:

$$AR = \frac{\alpha}{sd(\varepsilon_t)}. \tag{12}$$

Hecht (2014, p. 5) also cites a mathematical proof that shows that the maximum attainable SR of a total portfolio is a function of the AR of a new asset and the existing SR of the total portfolio. This implies that for portfolio optimization, a higher AR from a new asset translates to a higher SR for the total portfolio. Statistically, although the AR divides the annualized α by the annualized standard error of the residuals, a proper z-statistic divides the raw α by its standard error. We note that the correlation between the ARs and the z-statistics is 1.

We applied the factor model to all 163 competitor portfolios using the seven benchmark indices and one benchmark index (the S&P 500). The motivation behind using the one-factor model is that competitors only had access to investable assets in the S&P 500 during the competition, and this represents a fair comparison. However, we also include the seven benchmark indices as factors in a seven-factor model to check for robustness to the broader market, which includes strategies such as equity long–short (LS) that competitors could have replicated. We note that using seven benchmark indices may be arbitrary, resulting in different α, p-values, or AR values. However, we test for robustness between the two models and note that modern investors now have many potential benchmarks accessible via low-cost mutual funds or ETFs, and the hurdles for active managers are higher. Nevertheless, we apply both models to compare the results.

Table 4 presents the results for the seven-factor model. Of the 163 αs in the seven-factor model, 12 αs were statistically significant at the 0.05 level, including the highest (annualized) α of 30.8%. However, 11 of the 12 statistically significant αs were negative. For comparison, the equal-weighted portfolio in the M6 competition had an α of 3.4% with a p-value of 0.26 and an AR of 1.16.

Table 5 presents the results for the one-factor model using the S&P 500 only. αs are similar to those for the seven-factor model. Of the 163 αs in the one-factor model, five were statistically significant at the 0.05 level, including the highest (annualized) α of 30.0%. However, four of the five statistically significant alphas were negative. For comparison, the equal-weighted portfolio in the M6 competition had an α of 3.1% with a p-value of 0.47 and an AR of 0.73.

2.2. Autocorrelation, R^2, and the joint test of equality

To check the autocorrelation of the competitors' returns, we apply a Durbin–Watson test and find that 20 of 163 competitors exhibit statistically significant autocorrelation. We use the R package prais (version 1.1.3) and apply the Prais–Winsten estimator to estimate the coefficients of the one-factor model in the presence of autocorrelation. Results are fairly similar and there are still five statistically significant αs, with four of five αs negative. However, the original competitor with the highest α value keeps a similar α when rounded (30.0%), but its p-value shifts from 0.041 to 0.0501. Additionally, a new competitor with a positive α value of 7.4% (rounded) became significant, with its p-value shifting from 0.065 to 0.035 after the autocorrelation correction. Overall, the one-factor model is fairly robust to the autocorrelation in the competition dataset.

We also compare the R^2 values of the one-factor model with the competitors' SR and α values. We find that those competitors with extreme α (and high values of SR) all had R^2 values below 50% (and frequently below 20%). The 95% confidence interval for the correlation between the absolute value of α and R^2 was $(-0.48, -0.22)$, indicating that higher values of R^2 were associated with α closer to 0. The same held true when R^2 was compared with the SRs, which are not dependent on the output

Table 4
Summary statistics of annualized alphas, annualized ARs, and coefficients for the seven factor models.

	Value-add			Factor model coefficients						
	Annualized α[a]	p-value	Appraisal ratio[b]	S&P 500	NASDAQ-100	Crypto	CTA	Equity LS	Event-driven	Market-neutral
Min.	−0.41	0.00	−3.23	−2.23	−1.47	−0.05	−0.82	−2.86	−2.60	−2.31
1st Qu.	−0.07	0.21	−1.07	0.05	−0.30	−0.01	−0.14	−0.22	−0.20	−0.31
Median	−0.02	0.37	−0.25	0.39	−0.16	0.00	0.03	0.29	0.16	−0.07
Mean	−0.03	0.44	−0.24	0.35	−0.14	0.00	0.07	0.35	0.13	0.02
3rd Qu.	0.03	0.69	0.66	0.63	0.00	0.01	0.23	1.06	0.38	0.33
Max.	0.31	1.00	2.16	2.70	1.28	0.05	1.27	5.13	2.82	2.53

[a] The factor model estimates of α on a daily level. αs are annualized by multiplying the intercept by the number of trading days (238) in the competition, but the industry standard is normally 252 days.
[b] ARs are annualized by dividing the annualized αs by the annualized standard deviation of returns.

Table 5
Summary statistics of annualized αs, annualized ARs, and coefficients for the one-factor models.

	Value-add			Factor model coefficient
	Annualized α	p-value	Appraisal Ratio	S&P 500
Min.	−0.42	0.00	−3.19	−0.76
1st Qu.	−0.06	0.29	−0.87	−0.01
Median	−0.01	0.47	−0.16	0.32
Mean	−0.02	0.51	−0.21	0.31
3rd Qu.	0.03	0.75	0.49	0.68
Max.	0.30	0.99	2.06	1.32

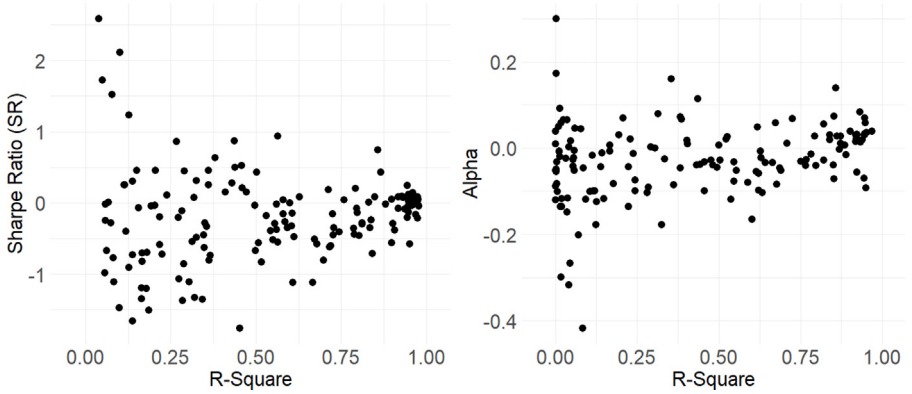


Fig. 4. Scatterplots of R^2 vs. SR and α, using a one-factor (S&P 500) model.

of a factor model. The 95% confidence interval for the correlation between the absolute value of the SR and R^2 was (−0.62, −0.41), indicating that higher values of R^2 were associated with SRs closer to 0. Overall, these results imply that competitors who performed at the extremes (i.e., won the competition) tended to be less dependent on the benchmark index. Similar findings held true when the seven-factor model was used. Fig. 4 displays the results for the one-factor model.

Finally, we perform a Gibbons, Ross, Shanken (GRS) test (Gibbons et al., 1989) for the joint equality of all α being equal to 0. The null hypothesis is

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_{163} = 0,$$

and it is rejected when there is at least one α not equal to 0. We implement the GRS test using the R package GRS.test (version 1.2) for both the one-factor and seven-factor models. For the one-factor model, the test statistic is 15.3 with a p-value < 0.0001, rejecting the null hypothesis that all α are equal to 0. For the seven-factor model, the test statistic is 15.1 with a p-value < 0.0001, rejecting the null hypothesis that all α are equal to 0. Since both tests rejected their null hypotheses, we conclude that at least one of the competitors' α is different from 0. However, we provide this conclusion with caution because the GRS test may have higher rejection rates when the portfolios are non-normal (Affleck-Graves & McDonald, 1989). The M6 Competition had more portfolios across a shorter time period (163 portfolios over 1 year) than Affleck–Graves and McDonald (1986; 20 portfolios over five years), which may further increase rejection rates. In this study, the high dimensionality of the covariance matrix (163 × 163) relative to the sample size (238) may lead to instability of its inversion when the test statistic and spurious rejections are computed. Furthermore, the GRS test tends to have extreme power, which means the null hypothesis can be rejected even when the deviation from the null is economically negligible (Kim & Shamsuddin, 2017).

2.3. Theoretical commentary on the maximum drawdown

As mentioned, investors use the MDD to manage risk and influence decision-making. The MDD calculates the maximum loss an investor experiences over time and is a maximal order statistic. It is the maximum percentage loss over any $m \leq n$ consecutive trading days. In our case, we assume that the returns from each of the n trading days in the competition are i.i.d., which we believe is a realistic assumption. Thus, the percentage loss of m consecutive days can be represented as one minus the product of m consecutive returns, and its maximum is a simplification of Eq. (4) when the m days occur from t_1+1 to t_2.

Theoretically, for maximal order statistics, if there are n observations that are i.i.d. with a common cumulative distribution function F, then the behavior of any middle order statistic j is rather simple and straightforward. Its mean is about $F^{-1}(j/(n+1))$ with variance of order $1/n$. However, the distribution of the extreme order statistics, say the largest, can be complicated, as it depends on the common distribution F. For the standard normal distribution, its mean is of order square root of $log(n)$ and variance is of order $1/log(n)$. For the exponential distribution, the mean is of order $log(n)$ with finite variance. For distributions with Pareto tails, both the mean and the variance grow algebraically with the sample size n. These results point to two aspects of the largest order statistic, such as the MDD. First, the mean grows as the sample size grows. Second, the largest order statistic is unstable, as its variance can decrease to zero slowly (normal case), or to a non-vanishing constant as in the exponential case, or slowly explode as in the Pareto case. A good account of these results can be found in David and Nagaraja (2003). For stationary sequences, the behavior of the middle order statistics is stable, as in the i.i.d. case. However, extreme order statistics can exhibit unstable behavior, and a good deal of technical work is available, such as Hsing (1988).

3. Momentum strategies

Sometimes, a team of high-performing competitors may underperform a carefully selected competitor team. Burghardt et al. (2007) found that a team of uncorrelated portfolio managers outperformed a team of portfolio managers with the highest SRs (Burghardt & Walls, 2011). This decision problem is common in the fund of (hedge) funds (FoFs) space, where a FoF's manager decides allocations of investments to several individual portfolios or hedge funds. Interestingly, they also found that "firing team players for poor individual performance and replacing them with managers with higher Sharpe ratios seriously degraded the performance" (Burghardt et al., 2007, p. 1).

To test this scenario in the M6 Competition, we create two strategies for FoFs seeking to invest in M6 Competitors. The first strategy picks the top 10 performing competitors (Superstars) from the prior month for an equal-weighted investment during the next month. The second strategy picks the bottom 10 performing (Superlosers) competitors from the prior month for an equal-weighted portfolio during the next month.

FoFs typically reinvest quarterly because there are fixed costs for switching between portfolios. We employ a FoF approach to simulate this exercise but assume monthly reinvesting at the M6 Competition selection dates to ensure a synchronized and fair comparison with the competition. For realism, we also assume that the competitors' underlying positions were unknown, and we only allow direct investments to competitors. We also do not permit competitors to charge management and performance fees, since the effects were minimal in this short competition.

3.1. Superstars strategy

Fig. 5 shows the NAV of the top 10 superstars during each month of the competition, where we assume that each competitor starts with \$100 each month. The figure also plots an average of the top 10 superstars (bold font, "Superstar Average") with the realized returns from the previous month. The bold line resets to a NAV of \$100 each month. The team of Superstars is selected before the selection dates of the competition by choosing the 10 competitors with the highest returns from the previous month (gray lines in the figure). This process starts in month two and is repeated for every subsequent month. The blue line represents the Superstar strategy and shows a steady decline in the NAV. We find an ending NAV of \$87.46 in the Superstars strategy, indicating mean reversion behavior of the prior month's Superstars.

3.2. Superlosers strategy

The process is repeated for the Superlosers strategy, except that the Superlosers consist of the bottom 10 performing competitors of the previous month. Fig. 6 shows an ending NAV of \$106.95 using the Superloser strategy, indicating that the prior month's top losers tend to increase their NAVs in the next month.

The unexpected performance of Superlosers vs. Superstars presents a challenge for investors who manage FoFs or invest in active investment managers. Many investors who engage in or believe in active investment are familiar with the concept of momentum investing and may apply such an approach to investing in managers. While momentum has been described as a potentially successful strategy for stocks (Jegadeesh & Titman, 1993), futures contracts (Moskowitz et al., 2012), broad asset classes via long-only investment in ETFs (Faber, 2007), and almost everything that is liquid and tradeable (Asness et al., 2013), it does not appear to extend to active investment strategies.

The performance of the Superloser portfolio does not, on its own, suggest that allocation to active strategies is a simple manner of buying the worst performers each month, and even if one desired to apply such a strategy, it is not practical in many cases. While ETFs and mutual funds may be traded in and out of frequently (barring trading restrictions in mutual funds), most hedge funds do not allow frequent trading and often only allow investors to add, subtract, or redeem in entirety on a quarterly basis. Most hedge funds also require a minimum of one month's notice before making a change.

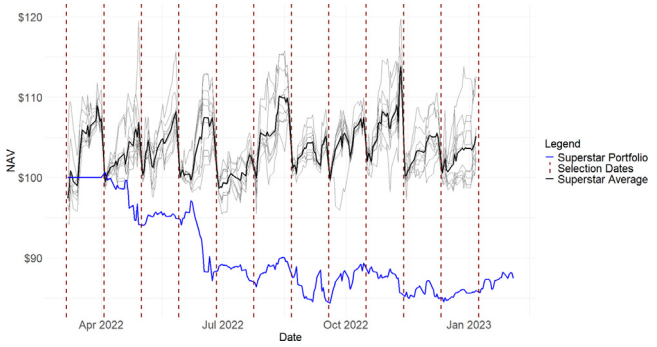


Fig. 5. Superstars strategy: investing in the top 10 winners of the prior month.

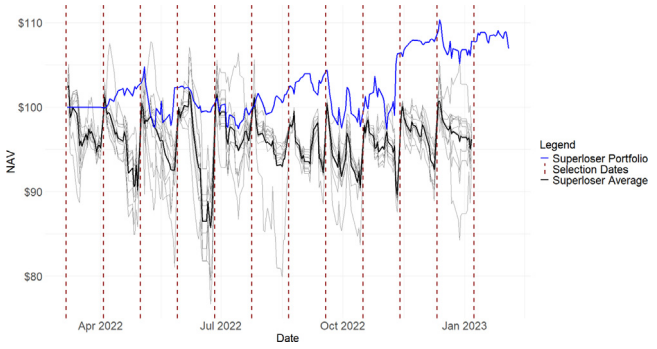


Fig. 6. Superlosers strategy: investing in the bottom 10 losers of the prior month.

4. Conclusions

M6 competitors were incentivized for extreme performance to win part of the $300,000 prize pool. They held long and short positions on 100 assets selected by the organizers from a clustering solution of average stock price, coefficient of variation, standard deviation, volume, and total returns. We found that this pool of assets performed better than industry-standard benchmarks such as the S&P-500 and the Nasdaq-100. We also found that the short competition made it difficult for competitors to demonstrate a positive value-add above industry-standard benchmarks, but most of the significant competitors had negative value-adds. Furthermore, competitors that had returns more similar to benchmark indices were less likely to demonstrate extreme performance in terms of SR (or IR). Overall, the winners of the M6 Competition were fairly robust to the choice of risk-adjusted financial metric used, whether it was the competition's IR or the additional risk-adjusted metrics introduced in this paper.

The M6 competition highlighted multiple challenges that face investment managers and investors who seek to allocate to active investment managers. For investment managers, it is quite difficult to beat portfolios composed of allocations that track simple equal-weight allocations to broad market indices, even over a short period, with a small universe representing only a fraction of the possible assets available. For end investors, it is very difficult to identify skill amongst investment managers. Perhaps more importantly, we see direct evidence that picking the managers with the best recent performance, an approach that echoes that of both retail traders as well as the momentum anomaly present in almost every liquid asset class, appears to be a recipe for underperformance.

The objective of winning a short-term (1-year) competition differs from the objectives of long-term investment managers in practice. For short-term competitions, competitors need to differentiate themselves to increase their chances of winning (Lichtendahl & Winkler, 2007). There is no financial downside for a competitor taking a large loss. For long-term investment managers, success requires maximizing fees over a career, which improve when large losses are avoided and clients/assets are maximized. They are expected to deliver a small yet positive tracking error relative to their benchmark and have a responsibility to their investors to minimize all kinds of risk (e.g., financial, legal, counterparty, key man, headline). To improve the real-world applicability of financial forecasting competitions, we offer a few suggestions for the future of forecasting competitions:

1. *Give competitors skin in the game*: Schneider et al. (2024) suggest starting competitors with initial capital to fictionally trade within the competition. Real

money gives competitors skin in the game, so that they have something to lose when taking large risks. However, we suggest augmenting this with a pre-filtering stage to select competitors that will act in good faith and actively manage their investments during the competition. Fictional losses and the extra prize pool money could then be allocated to the winners.

2. *Include random portfolios as a benchmark*: In 1973, Burton Malkiel infamously declared, "A blindfolded monkey throwing darts at a newspaper's financial pages could select a portfolio that would do just as well as one carefully selected by experts" (Malkiel, 1973). Can a forecasting competition distinguish sustainable good performance from luck? We suggest future forecasting competitions include random portfolios as an additional benchmark to an equal-weighted portfolio. Competitors' performances can be compared to quantiles of the distributions of random portfolios, provided those random portfolios are properly constructed (see Burns, 2007; Kim & Lee, 2016; Lisi, 2011; Stein, 2014).
3. *Allow Fund-of-Funds (FoFs) competitors to compete*: FOFs competitors could be introduced and allowed to invest in other competitors. A FoFs competitor might decrease their risk by allocating their portfolio to a subset of 10 competitors that were uncorrelated (or negatively correlated) to each other. These FoFs competitors may produce winning strategies by analyzing the uniqueness of competitors' returns or avoiding investments in competitors which behave similarly. Individual competitors with desirable statistical properties could also be rewarded for being selected by the FoFs competitors. For valuing individual competitors, see the work by Raja et al. (2024) on aggregation operators, scoring rules, and payoff mechanisms.
4. *Highlight the losing strategies*: Nearly all forecasting competitions, including the M Competitions, highlight the strategies of winning competitors. However, competitions are short-lived, and investment managers want to know what strategies to avoid in the long-term. Thus, future competitions should include explanations for the performance of the competition losers. A future competition could also award a prize to a competitor that minimized all forms of risk (e.g., drawdown, ulcer index, downsize volatility, annualized standard deviation, VaR, Expected Shortfall) while only achieving a minimum benchmark return.

The authors appreciate the opportunity to review the results of the competition and offer our congratulations and respect to the organizers and participants.

CRediT authorship contribution statement

Matthew J. Schneider: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Rufus Rankin:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Prabir Burman:** Writing – original draft, Methodology, Formal analysis. **Alexander Aue:** Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.03.008.

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